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AHMAD WIDARDI

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Summary

Principal Software Development Engineer with 10 years of experience building scalable enterprise software and AI platforms, including 4+ years in technical leadership and engineering management. Currently leading and architecting core components of the **GLChat AI ecosystem**—specializing in web crawling pipelines, automated data ingestion, intelligent enterprise search, and LLM-powered deep research orchestration. As an **AI-native engineer**, I apply modern agentic practices across the development lifecycle, including prompt engineering, context engineering, tool and model orchestration, iterative agent loops, and evaluation-driven development. Deep backend and distributed systems expertise across **Java/Spring Boot** and **Python/FastAPI**, complemented by real-time computer vision engineering using **OpenCV**. Proven track record of architecting large-scale enterprise chat platforms for **Bank BCA** (**Halo BCA Chat**, **Winters**, **HCI Inspire**) supporting customer service operations, nationwide branch employees, and HR users, with **Halo BCA** serving **40M+** active users. Grounded in a strong algorithmic foundation as a former **ACM-ICPC Asia Regional Finalist** and **International Mathematics Competition Bronze Medalist**.

Work Experience

[GDP Labs] Principal Software Development Engineer II (January 2026 - now)

- **Engineering Leadership:** Serve as **Team Lead** guiding project pods of 6 engineers, managing and mentoring 7 team members across high-stakes data infrastructure deliverables
- **Technologies:** Crawl Pipeline, Web Search Pipeline, Deep Research, Firecrawl, Playwright, Puppeteer,Google Connector, Microsoft Connector, Github Connector, Tongyi, SeaLion, GPT-Researcher, PostgreSQL**.**
- **Frameworks:** Python Fast API, Python Fast MCP**.**
- **Core Products**
  - **GL Smart Crawl (Automated Data Ingestion):** Architected an incremental web scraping microservice (**Python**) as a shared GL ecosystem capability. Engineered continuous source monitoring and structured extraction pipelines to feed raw web data into downstream data lakes via dedicated indexes.
  - **GL Smart Search (Unified Enterprise Search Ecosystem):** Led development of a unified text-based search platform (**Python**). Engineered robust integrations to aggregate and index data across the live web, internal knowledge bases, **Google Workspace** (Drive, Mail, Calendar), **Microsoft 365** (Outlook, OneDrive), and **GitHub**.
  - **GL Open DeepResearch:** Built an automated, task-based research orchestrator (**Python**) unifying open-source deep research libraries under a single API, integrating advanced models (**Tongyi**, **SeaLion**, **GPT-Researcher**).

[GDP Labs] Principal Software Development Engineer I (April 2022 - December 2025)

- **Engineering Leadership:** Serve as **Team Lead** guiding project pods of 6 engineers, managing and mentoring 10 team members across high-stakes data infrastructure deliverables.
- **Technologies:** Web Search Pipeline, Admin and Configuration Pipeline, Deep Research, Open CV (Java), Firecrawl, SWIRL, Stack Auth, Directus, Supabase, AWS S3, Elasticsearch Vector Search, Redis, Chroma, FFmpeg, PostgreSQL.
- **Frameworks:** Python Fast API, Python Fast MCP, Spring Boot, Nest JS, Next JS, React JS, Ejabberd.
- **Core Products - GLChat Enterprise AI Platform:** Contributed to the development of **GLChat**, an enterprise multi-tenant B2B **Generative AI platform** (**Python**, **Next.js**) designed to help organizations securely build, deploy, and manage domain-specific AI chatbots and agents using their internal knowledge bases. Contributed in components:
  - **GLChat Backend:** Contributed to backend API implementation and integrated GLChat authentication (**Stack Auth**). Spearheaded the architectural design and migration to move environment configurations from **Excel** into the dedicated **GLChat Admin Dashboard**.
  - **GLChat Admin Dashboard**: Served as **Team Lead** and built administrative management interfaces (**Python**, **Next.js**) to govern tenant management, chatbot configurations, PII masking parameters, tool access, and dynamic role-based user controls (excluding infrastructure management).
  - **GL Smart Search:** Served as **Team Lead** for **search pipelines** within GLChat. Built **Web Search**, **Connector features**, and an early-stage **Deep Research** feature (later became **GL Open DeepResearch**) . Engineered a dynamic FAQ generation pipeline utilizing **KNN-based candidate retrieval** and **Farthest Point Sampling (FPS)**.
  - **GL SDK (Maintenance):** Maintained core libraries, specifically overseeing **GLLM Datastore** (Elasticsearch vector DB, Chroma, Redis) and **GLLM Retrieval** components.
- **Enterprise Projects (Revenue Generation)**

- **Halo BCA Chat Platform:** Bank BCA's flagship enterprise customer support system powering real-time chat routing, automated agent distribution, and back-office analytics for **40M+ active users** handling heavy payday traffic surges.
- **Halo BCA Chat Database Housekeeping:** Drove the complete lifecycle—from gathering requirements directly from Bank BCA executive stakeholders, to system architecture and technical execution—for an automated database housekeeping application. Safely archived and managed over **600 GB** of decade-long chat history and session data while ensuring zero-downtime data migrations capable of processing **1 million rows per peak day**.
- **Pertamina Vehicle Queue and Working Area Compliance Detection**: Architected real-time CCTV video stream processing engines using **OpenCV (Java)**, **Spring Boot**, and **AWS S3** to automate vehicle and employee detection for work-area safety compliance, including automated alert mailing workflows.
- **Savoria SIRE (High-Performance Search & Forecasting)**: Led Phase 1 full-stack development (**NestJS**, **Next.js**) and resolved critical inventory lookup bottlenecks by introducing **Elasticsearch**, significantly accelerating query performance and refactoring complex sales forecast models.
- **BCA Digital Greeting (Automated Multimedia Generation)**: Designed and executed an automated video rendering pipeline utilizing **Spring Boot** and **FFmpeg** to dynamically generate personalized greeting videos (Birthdays, Christmas) for **Bank BCA Priority** and **Solitaire** customers.
- **BCA Data Property Crawling (Headless Data Crawling Engine)**: Architected a resilient property data extraction backend (**Next.js**) utilizing **Puppeteer** (headless browser) and **Spring** proxy routing to scrape and aggregate market data across major property platforms (e.g., rumah123), integrating directly with **GL Smart Search** APIs.
- **Maintenance Internal BCA Platforms and PING**: Executed core enhancements and Microsoft Teams integrations for **Winters**. Maintenance support for the **Halo BCA Mobile SDK** (**Android/iOS**), **HCI Inspire** (HR platform), and the **PING** messaging routing engine.
- **Yamaha Music - YMMI (AI Inspection Box & Self-Hosted OCR)**: Architected and developed an AI Inspection Box portal (**Spring Boot**, **React.js**) enabling automated scanning, OCR verification, and web portal registration of product codes on physical boxes.
- **Winters BCA**: Served as **Team Lead** & **Architect** for Bank BCA's nationwide internal CS platform, connecting branch employees across Indonesia with central Jakarta operations, directly driving a **major** share of GDP Labs' overall corporate revenue.
- **HC Inspire**: **Team Lead** for Bank BCA's internal HR support platform, solving architectural blockers to restore nationwide availability for all BCA employees, directly driving a **minor** share of GDP Labs' overall corporate revenue.
- **PING (Real-Time Messaging Infrastructure)**: Maintained RESTful microservices using **Spring Boot** for GDP Labs' proprietary messaging engine (**PING**), processing incoming real-time chat payloads and asynchronously forwarding routing requests to **Ejabberd** (**XMPP**) messaging servers.

[GDP Labs] Senior Software Development Engineer (November 2020 - March 2022)

- **Technologies:** Java 8, Spring Boot, Python, PostgreSQL, Oracle DB, Elasticsearch, GraphQL Subscriptions.
- **Backend Architecture & Real-Time Forex Trading (BCA Treasury JValas):** Architected Phase 1 microservices using **Java 8**, **Spring Boot**, and **Oracle DB**. Integrated **Refinitiv Trading APIs** and **GraphQL Subscriptions** (WebSockets) for real-time forex streaming, successfully guiding executive UAT.
- **Feature Enhancement and Performance Improvement (Halo BCA Chat):** Resolved **multi-million** record **PostgreSQL** bottlenecks by migrating to an **Elasticsearch** query engine and shipped core feature enhancements.
- **Winters:** Served as **Team Lead** & **Architect** to build and launch the initial production deployment.
- **HCI Inspire:** Contributed in backend refactoring and resolved critical production deployment outages.
- **Architecture & Development (GLAIR AI Licensing Platform):** Designed, architected, and built the end-to-end **License Management System (LMS)**; engineered a hybrid **Spring Boot**, **Python**, and **PostgreSQL** platform to issue, validate, and manage software licensing for modular **GLAIR AI** microservices across online and offline deployments.
- **Engineering Leadership & Mentorship**: Created GDP Labs' official **SDE Onboarding Framework**, standardizing technical training and accelerating time-to-productivity for incoming software engineers.

[GDP Labs] Software Development Engineer (September 2018 - October 2019)

- **Technologies:** Spring Boot, GraphQL, S3, Ejabberd (XMPP).
- **Refactor (Halo BCA Chat):** Refactored core **Halo BCA Chat Backend**, upgrading **Spring Boot** frameworks, integrating a file management microservice using **S3-compatible** presigned URLs, and implementing **GraphQL** APIs.
- **Real-Time Messaging Infrastructure (PING)**: Minor enhancements in **PING**.
- **Code Review Project (BCA Blu Backend)**: Performed technical code reviews for **Bank BCA digital banking** consumer microservices, identifying critical transaction atomicity and data consistency issues to ensure platform reliability.

- **Quality Assurance & Performance Guidelines**: Co-authored **GDP Labs**' internal **Software Quality Guideline** , establishing standardized performance testing procedures and quality assurance frameworks across engineering teams.
- **Technical Hiring & Assessment**: Authored competitive programming problems and configured automated testing environments to streamline technical candidate screening and recruitment.

[Blibli.com] Software Development Engineer (September 2018 - October 2019)
- **Technologies:** Spring MVC, Spring Boot, MongoDB, PostgreSQL, Apache Kafka, Redis, React.js, Vue.js, Angular.js.
- **Core Backend & Microservices:** Engineered high-throughput microservices powering **Main Commerce Workflows**, **Campaign**, **Promotion**, and **Cart** services using **Spring MVC**, **MongoDB**, **PostgreSQL, Apache Kafka, and Redis.** Built front-end components using **React.js**, **Vue.js**, and **Angular.js**, connecting them to RESTful back-end services.

[Blibli.com] Blibli Future Program (29 July 2016 - February 2018)
- **Career Page Revamp:** Revamped **Blibli's external career site** and **internal hiring system** using **Spring Boot** and **Vue.js**.
- **Commerce Center:** Built the back-office portal from scratch using **Spring Boot** and **Angular.js**.

Education

Yogyakarta, Indonesia **Gadjah Mada University** **September 2019–July 2022**
- Master: Computer Science Major. Cumulative GPA: 3.57 (4 semester).
- Thesis : Haar Classifier Method to Detect Human Position on Non – Overlapping CCTV Camera
  - Link : https://bit.ly/thesis-master-awidardi

Yogyakarta, Indonesia **Gadjah Mada University** **September 2014–July 2018**
- Undergraduate: Computer Science Major. Cumulative GPA: 3.52 (8 semester).
- Thesis : Page Dewarping on Text – Based Image by Using Pixel Addition Method
  - Link : http://bit.ly/thesis-awidardi

Selected External Projects

- **Paperplane (2022 - 2024) -** A school – assistant product which focuses on developing curriculum for school from student selection until the student is ready to be an Olympiad student. Developed with **Nest JS** and **React JS**.
- **Arisan Tumbuh (2020) -** Arisan Tumbuh is an **MNC Hackathon 2021 winning application** that enables users to do ROSCA / *arisan* activities through mobile apps without losing its social aspects and characteristics.
- **Pena Kita (2018) -** An award-winning innovation that provides a public administration solution for Indonesian citizens that live in 3T (very poor) areas. **PENA KITA** was awarded a **GOLD MEDAL** in **GEMASIK 11** and **Outstanding Student Award** (Insan Mahasiswa Berprestasi) 2018 by Universitas Gadjah Mada.

Additional Awards

Awards
- **Bronze Medal and Merit Award International Mathematics Competition Incheon, South Korea (2010)**
- **Gold Medal GEMASTIK 2018 by Ministry of Research and Higher Education Indonesia**
- **Insan Mahasiswa Berprestasi National Gold Medalist (Outstanding Students Award) by Universitas Gadjah Mada 2018**
- **1st Winner MNC National Hackathon 2020**
- **3rd Winner @america Legal Tech Hackathon 2019**
- **ACM - ICPC Regional Contest Honourable Mention (2019 Asia Danang; 2016,2015,2014 Asia Jakarta; and 2015 Asia Singapore) . ICPC ID: Certificate**
- **Bronze OSN Mathematics SD 2006**
- **Finalist OSN Mathematics SMP 2009 and SMA 2013**
- **4th Winner National Training (Pelatnas) IMC 2010**
- **Grand Finalist (6th Place) for Cyber Jawara, National Cyber Security Competition 2018 at Bali, Indonesia**
- **Finalist (6th Place) for Networking and Cyber Security GEMASTIK 2017**
- **Finalist for Software Development (PPL) GEMASTIK 2016**

Non Technical Awards
- **GDP Labs Employee of the Month January 2022 and February 2022**

Sidiq Tri Pratikto

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Summary

Senior Software Engineer with 5+ years of experience across Android/iOS development, backend APIs and microservices, and AI agents. Strong algorithmic foundation: Codeforces Master (peak rating 2180) and ICPC Asia Regional medalist.

Experience

GDP Labs Nov 2021 – Present
Software Development Engineer → Senior Software Development Engineer *Yogyakarta / Jakarta, Indonesia*

GL Consulting (client placement) – Android Engineer, then AI Agents / Digital Employee Engineer *2025 – Present*
- Build AI agents for a "Digital Employee" Project Manager, including a minutes-of-meeting (MoM) assistant and information storage & retrieval.
- Developed Android features for client projects before moving to the AI agents team.

GLAIR (client placement) – Senior Software Development Engineer, Mobile *Jul 2024 – 2025*
- Shipped features and fixed bugs on iOS and Android apps for CCEP AUNZ SaM M and CCEP ID Roam Sales, working with cross-functional teams.
- Improved user experience by resolving existing bugs and optimizing app performance.

KASKUS Networks (client placement) – SDE, promoted to Senior SDE in Nov 2022 *Jan 2022 – Jul 2024*
- Developed KASKUS Next API; built an API that synchronizes video duration from a microservice for KASKUS TV.
- Automated live reports for KASKUS TV, improving data-retrieval efficiency; built content moderation for Multimedia Threads.
- Increased test coverage and supported the data team with GDS reports.

GDP Labs internal – Coding Test Task Force *2021 – 2022*
- Authored and reviewed coding-test problems for SDE and AIE recruitment; analyzed Coding Test 2022 results to identify top candidates.

GDP Labs Nov 2018 – Jan 2019
Software Development Engineer Intern *Yogyakarta, Indonesia*
- Explored Facebook Pixel and WebRTC, then implemented both on a local site.

Tokopedia Dec 2016 – Mar 2017
Software Engineer Intern *Jakarta, Indonesia*
- Integrated Pinpoint Validation v2 on the mobile site for on-demand logistics (Gojek, Grab, etc.).

Education

Universitas Gadjah Mada 2014 – 2021
Bachelor's degree, Computer Science *Yogyakarta, Indonesia*

Competitive Programming & Awards

Ratings: Codeforces Master (max 2180, 1,300+ problems solved); TLX 2077; AtCoder 2-kyu (1619).
ICPC: Asia Jakarta 2018, National Bronze Medal; Asia Singapore 2018 and Asia Da Nang 2019, Honorable Mention; Asia Jakarta 2015, Finalist (Top 50).
National: Gemastik 2018 Programming, Finalist (Top 12); Kalibrr Kode Indonesia 2017, Top 10 of 5,000+ contestants.
Regional: Techphoria 2018, Silver; Ideafuse ICPC 2018, Bronze; Porsematif UII 2016, 3rd; Bukalapak Programming Contest 2016, 5th; Petra ICPC 2016, 5th; Ideafuse ICPC 2015, 5th.

Technical Skills

Languages: Kotlin, Java, Swift, Python, C++, PHP, SQL
Domains: Android and iOS development, REST APIs and microservices, AI/LLM agents, algorithms and data structures
Tools & Cloud: Git, AWS

Certifications & Languages

Certifications: AWS Partner Accreditation (Technical), 2022; HackerRank Problem Solving (Advanced), 2020; HackerRank SQL (Basic), 2022
Languages: Indonesian (native), English

ERWIN EKO WAHYUDI

PERSONAL DATA

Name : Erwin Eko Wahyudi
Date of Birth : October 28th, 1994
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Phone : (+62) 812 1195 4011
Email : erwin.eko.w@ugm.ac.id
Affiliation : Department of Computer Science and Electronics, Faculty of Mathematics and Natural Sciences, Universitas Gadjah Mada, Indonesia



EDUCATION

M.Cs Graduate Program of Computer Science, Universitas Gadjah Mada
August 2017 - April 2019
Thesis: "On the Effect of Splitting Data in Extreme Learning Machine"
Supervisor: Dr.-Ing. Reza Pulungan
GPA : 3.90 (Cumlaude)

S.Kom Undergraduate Program of Computer Science, Universitas Gadjah Mada
August 2012 - February 2017
Thesis: "Counterexample Generation for Ping-Pong Protocol Security Checking Algorithm"
Supervisor: Dr.-Ing. Reza Pulungan
GPA : 3.81 (Cumlaude)

WORKING EXPERIENCE

Lecturer, Universitas Gadjah Mada 2021 - now

Artificial Intelligence Research Engineer, Bukalapak 2019 - 2021
Involved in a personalized ranking project as well as deploying and maintaining a large-scale recommender system

Instructor, Jogja Science Training 2016 - 2019
Preparing students for Mathematics Olympiad and Informatics Olympiad

HONORS AND AWARDS

Honourable Mention of The 23rd International Mathematics Competition (IMC) for Undergraduate Students 2016
Mathematics Olympiad held in American University in Bulgaria

First Winner of Bukalapak Programming Contest Yogyakarta 2016
A competitive programming competition held by Bukalapak

Third Winner of Ideafuse ACM ICPC Provincial Contest 2016
A competitive programming competition held by Mikroskil

Gold Medalist of National Mathematics Olympiad for Undergraduate Students 2015
National Mathematics Olympiad held by The Ministry of Education

First Winner of Lumia Apps Olympiad 2012
Displaying the catalogue of KrazyMarket in a Nokia Lumia application

TEACHING EXPERIENCE

August - December 2022
- Fundamentals of Linear Algebra
- Programming
- Programming Lab Session

February - June 2022
- Algorithms and Data Structures
- Algorithms and Data Structures Lab Session
- Software Engineering Workshop

August - December 2021
- Software Engineering Workshop
- Programming Lab Session

February - June 2021
- Discrete Mathematics
- Elementary Differential Equations
- Programming Lab Session

PUBLICATIONS

Journals

Wahyudi, E. E., Auzan, M., Dharmawan, A., Nuryanto, D., Susyanto, N., Samodra, G., & Hadmoko, D. (2022). Akuisisi Data Prediksi Curah Hujan Secara Periodik Menggunakan Apache Airflow (written in Bahasa Indonesia). *Journal of Informatics Information System Software Engineering and Applications (INISTA)*, 4(2), 1-12. https://doi.org/10.20895/inista.v4i2.574

Rokhman, N., Wahyudi, E. E., & Hendarto, J. (2022). The Solution of Non-Linear Equations System Containing Interpolation Functions by Relaxing the Newton Method. *ComTech: Computer, Mathematics and Engineering Applications*, 13(1), 45 52. https://doi.org/10.21512/comtech.v13i1.7322

Conference Papers

E. E. Wahyudi and R. Pulungan, "Counterexample Generation for Ping-Pong Protocols Security Checking Algorithm," 2018 6th International Conference on Information and Communication Technology (ICoICT), 2018, pp. 293-298, doi: 10.1109/ICoICT.2018.8528754.

PROFESSIONAL TRAINING

AWS Certified Cloud Practitioner
Issued by AWS, September 2022

Microsoft Azure AI Fundamentals
Issued by Microsoft, May 2021

COMMUNITY SERVICE

Instructor on Training of New Informatics Curriculum for High-School Teachers
Yogyakarta, October 2022

LANGUAGES

Indonesia: Native language

English: Limited working language